

INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY

Spatial informatics

Time:45 minutes

Total Marks: 25

Section-I (10 marks)

1. Mr. X has digitized a neighbourhood land-use map showing 10 land-use classes. When he examines the attribute table, he finds more than 10 polygons, including several very small polygons. What are these very small polygons called?
2. Which vector data model stores coordinate strings one after another without any internal topological structure?
3. Which process combines socio-economic data with other datasets to produce neighbourhood profiles and classify areas for marketing purposes?
4. Which instrument can measure both angles and distances with millimetre-level accuracy?
5. Which technique obtains information about the physical, chemical, and biological properties of objects without physical contact?
6. Which process converts line graphics into digital data by breaking continuous lines into a series of discrete points?
7. Which surface representation depicts the complete topography of the Earth's surface, including both natural and anthropogenic features?
8. What is the measurement and mapping of water depth and underwater terrain called?
9. An ecologist has collected sample data about individual plants in a forest and wants to generate a continuous surface map showing the spatial density of plants. Which GIS technique can be used for this task?
10. Which technique is used to obtain highly precise ground control points for geometrically correcting high-resolution satellite imagery or aerial photographs?

Section II (10 Marks)

11. A survey was conducted in a city to record pollution levels at different locations. Assuming an imaginary city boundary and suitable pollution values, how would you represent the spatial variation in pollution levels using GIS? Depict your answer in a suitable pictorial format.
12. Classify the following attributes according to their appropriate data types:
Phone numbers, Rainfall (mm), Population, Height of trees
13. List any two problems that may arise when converting a paper map into a digital map.
14. The figure represents a mixed-pixel situation. Identify the two methods used to assign a single class to each mixed pixel. Represent the resulting classified grid for each method.



15. The figure represents a raster image that needs to be compressed using Run-Length Encoding (RLE). Represent the given raster image using Run-Length Coding

	1	2	3	4	5	6	7	8
1								
2								
3								
4								
5								
6								
7								
8								

Section III (5 Marks)

16. Identify the topological relations in the following figure, write polygon, node, link topology and link coordinates

